

Discrete Mathematics Lecture 4 Predicate Logic

e.g. $\begin{cases} \text{Every human being is mortal} \\ \text{Socrates is human} \\ \text{Socrates is mortal} \end{cases}$ How can we use proposition to describe it? No!

So we need predicate logic.

x : a variable that represents an entity in domain.

$\text{Human}(x)$: x is a human, (this function transfer x to a proposition)

$\text{Mortal}(x)$: x is mortal.

$\forall$: a universal quantifier "for all". Maybe like " $\wedge$ ".

Every human is mortal: $\forall x (\text{Human}(x) \rightarrow \text{Mortal}(x))$

Socrates is a human: $\text{Human}(\text{Socrates})$

Socrates is mortal: $\text{Mortal}(\text{Socrates})$

$\exists$: the existential quantifier "there exists"

A Greek man discovered the Buoyancy: $\exists x (\text{Greek}(x) \wedge \text{buoyancy}(x))$.

$\left. \begin{array}{l} \text{Greek}(\text{Arc.}) \\ \text{Buoyancy}(\text{Arc.}) \end{array} \right\} \text{Greek}(\text{Arc.}) \wedge \text{Buoyancy}(\text{Arc.}) \rightarrow \exists x (\text{Greek}(x) \wedge \text{buoyancy}(x))$

predicates: a propositional function.

$P(x, y)$ is a statement, P is a predicate.

e.g. $P(x, y) := x < y + 1$, $P(1, 1)$ is T, $P(2, 1)$ is F.

$Q(n) := n > 2 \rightarrow n^2 > 4$ a tautology.

Summary: $x_1, \dots, x_n$: variables P : n -place function $P(x_1, \dots, x_n)$: proposition.

translation: $\forall x P(x)$: for all values of x in the domain

$\rightarrow$ negation: $\exists x, \neg P(x)$.

Question: translate: $\forall x_i P(x_1, \dots, x_n)$

$\left\{ \begin{array}{l} \forall x_i P(\dots, x_i, \dots) \text{ a predicate} \\ \forall x_i P(x_1, x_2, \dots, x_{i-1}, x_i, x_{i+1}, \dots, x_n) \text{ is a propositional function} \end{array} \right.$

translate: $\exists x (\text{Greek}(x) \wedge \text{buoyancy}(x))$

there exists at least one element x in the domain such that $P(x)$ holds.

Domain = $\emptyset$, $\forall x P(x)$: true $\exists x P(x)$: False

Remember: Domains are important.

Free and bounded variables.

x is bounded following a quantifier. else, it's free.

$P(x, y)$ will be a proposition when all the variable are bounded.

$P(x, y)$, x, y in $P(x, y)$ is the free variables, bounded one can't be asserted

Quantifiers with restricted domains.

e.g. $\forall x > 100, (x+1 > 101)$

How to eliminate conditions? $\forall x > 100 (x+1 > 101) \Leftrightarrow \forall x [(x > 100) \rightarrow (x+1 > 101)]$

$\forall x: Q(x, y), P(x, y) := \forall x (Q(x, y) \rightarrow P(x, y))$

$\exists x: Q(x, y), P(x, y) := \exists x (Q(x, y) \wedge P(x, y))$

e.g. $\lim_{n \rightarrow \infty} a_n = A$ $P(A) := \forall \epsilon > 0 \exists N \forall n > N (|a_n - A| < \epsilon)$

Connections: (with proposition).

$\forall x P(x) \Leftrightarrow \bigwedge_{i=1}^n P(v_i)$ $\exists x P(x) \Leftrightarrow \bigvee_{i=1}^n P(v_i)$

Logical Equivalence

Def: ϕ, ψ ; statements with predicates but without free variables.

if no matter which concrete predicates and domains are given, the truth value coincide

De Morgan's Laws $\left\{ \begin{array}{l} \neg (\forall x P(x)) \equiv \exists x \neg P(x) \\ \neg (\exists x P(x)) \equiv \forall x \neg P(x) \end{array} \right. \left\{ \begin{array}{l} \forall x P(x) \equiv \neg (\exists x \neg P(x)) \\ \exists x P(x) \equiv \neg (\forall x \neg P(x)) \end{array} \right.$

$\forall x (P(x) \wedge Q(x)) \equiv (\forall x P(x)) \wedge (\forall x Q(x))$ ✓

$\forall x (P(x) \vee Q(x)) \equiv (\forall x P(x)) \vee (\forall x Q(x))$ ✗

$\exists x (P(x) \vee Q(x)) \equiv (\exists x P(x)) \vee (\exists x Q(x))$ ✓

$\exists x (P(x) \wedge Q(x)) \equiv (\exists x P(x)) \wedge (\exists x Q(x))$ ✗